

German Grid Forecasts: Where the Uncertainty Bands Fail

A 13-month audit of German and Dutch day-ahead forecasts against what happened: systematic midday bias, uncertainty bands that fail out-of-sample (including conformal ones), and what that costs: $\approx$ €200,000 per 100 MW of avoidable imbalance exposure, concentrated in exactly the hours where failing is expensive.

KW-2026-01 · v1.0 analysis · 2026-08-17 · Matthew Tanti

ABSTRACT

We audit every quarter-hour of German (DE-LU) day-ahead system forecasts from July 2025 through July 2026 (load, solar, onshore and offshore wind) against realized outcomes from the ENTSO-E Transparency Platform. Three results. First, the point forecasts carry conditional bias that vanishes in aggregate statistics: load is over-forecast by roughly 1.9-2.0 GW every midday. Second, the two standard ways of wrapping intervals around a point forecast, parametric Gaussian bands and rolling split-conformal quantiles, both undercover out-of-sample (83-89% at nominal 90%), and collapse regionally: solar's nominal-80% band covered 56-63% through spring 2026. Distribution shift breaks the exchangeability that conformal's guarantee rests on. Third, adaptive conformal inference restores stated coverage by construction, at an honestly stated width cost. Priced at Dutch imbalance spreads (the NL leg of the audit), trusting the Gaussian "90%" band carried roughly EUR 362,000 of un-reserved adverse exposure per 100 MW over the period, versus EUR 158,000 under the adaptive band, with misses concentrated in the intervals where imbalance was most expensive.

INTRODUCTION AND NOTATION

the question · the data · notation

Every energy desk in Europe consumes the TSO day-ahead forecasts, and almost nobody asks the question this analysis answers: *when those forecasts imply a range of outcomes, is the range honest?*

We audit thirteen months of German (DE-LU) day-ahead forecasts against what actually happened (every quarter-hour from July 2025 through July 2026) for system load, solar, onshore wind and offshore wind: 38,016 quarter-hours per series, worst-case missingness 0.05%, all from the ENTSO-E Transparency Platform.

The findings run in increasing order of consequence: a conditional bias that averages conceal (§ 02); the failure, out-of-sample, of both standard ways to build uncertainty bands, including the naive conformal one (§ 03); and the adaptive construction that keeps its promise, at a stated price

(§ 04). § 05 records what this audit can and cannot show.

e	a forecast error: actual minus forecast, in MW; negative = the forecast was too high
Coverage	the fraction of the time an interval contained the outcome; a "90% band" claims coverage 90%
α	the miss rate a method aims for; a 90% band targets $\alpha = 10\%$
ACI	adaptive conformal inference: a feedback rule that adjusts α after each miss (§ 04)

§ 01 / 05

The data, and the rules of evidence

Public forecasts, public outcomes, one strict discipline: every number scored strictly out-of-sample.

The Transparency Platform publishes each TSO's day-ahead point forecasts alongside realized values. That pairing makes an audit possible without anyone's permission: the forecast was committed before delivery, the outcome is measured after it, and the difference (the error e = actual – forecast) accumulates into a track record nobody curates.

Two rules govern everything below. First, any quantity fitted from history (a mean, a standard deviation, a quantile) uses a trailing 90-day window that ends the day *before* the observation it is scored on. No method ever sees its own exam. Second, every construction is evaluated per quarter-hour delivery slot, because grid errors have a schedule: a method that is right on average and wrong every noon should be caught, not excused.

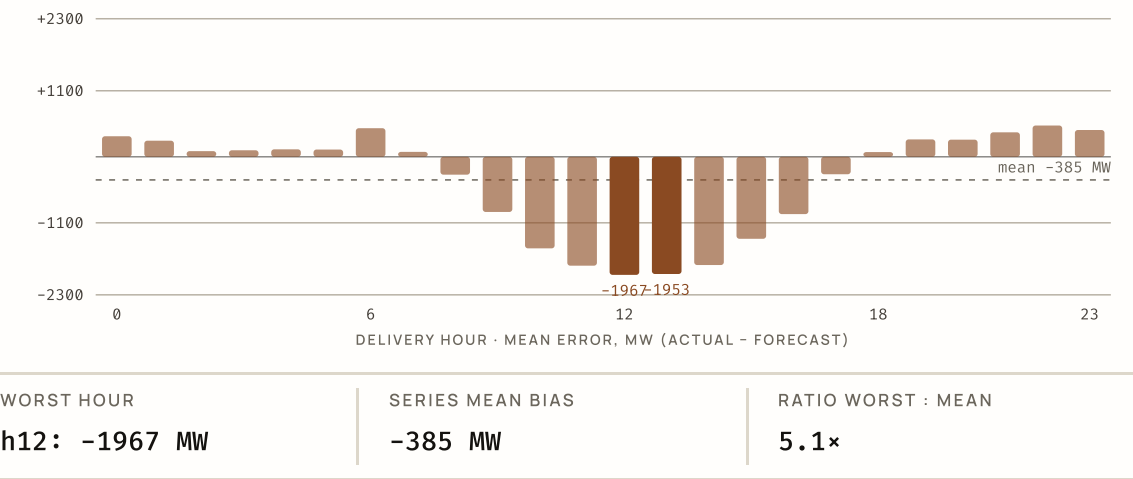
The bias clock

Germany's load forecast overshoots by two power stations' worth. Every day, at the same hours.

The aggregate numbers look respectable: load bias -385 MW on a ~55 GW system, solar -56 MW, offshore wind +176 MW. The clock tells another story.

FIG. 01: THE BIAS CLOCK

Mean forecast error by delivery hour (Europe/Berlin), 13 months of quarter-hours. Bars below the zero line mean the day-ahead forecast was *too high* at that hour. The overall mean (dashed) is what a summary statistic reports. The clock is what dispatch experiences.



ENTSO-E TRANSPARENCY PLATFORM • DE-LU • 2025-07-01 → 2026-07-31 • 38,016
QUARTER-HOURS/SERIES • MEAN ERROR BY DELIVERY HOUR

Load is over-forecast by **1.8–2.0 GW at hours 11–14**: roughly two large power stations of demand that is predicted daily and daily fails to materialize, five times the size of the aggregate bias. The likely mechanism is well known (behind-the-meter solar eroding metered load faster than the forecast model learns); the *size* is something you only learn by measuring. Offshore wind leans the other way, under-forecast by ~240 MW through the night hours; solar's midday under-forecast peaks around -420 MW.

PITFALL 2.1: THE AVERAGE FORGIVES

A summary bias statistic averages over the clock. A desk that trades the midday blocks does not. Any audit, internal or vendor, that reports a single bias number per series has not yet said anything about the hours where the money is.

Three ways to promise 90%

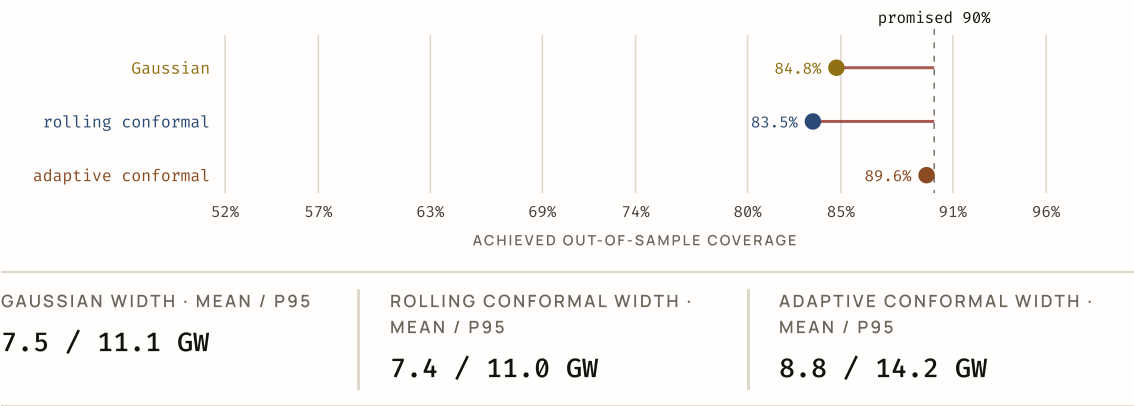
The parametric habit fails. So does the naive conformal alternative. Admitting that is the point.

The TSO publishes points, not ranges. Any desk wanting a range constructs it, and the standard construction is parametric: fit mean and standard deviation to recent errors, quote $\mu \pm z \sigma$. The textbook distribution-free alternative replaces the Gaussian fit with trailing empirical quantiles: a rolling split-conformal band. We ran both, identically windowed, strictly out-of-sample; and a third method introduced in § 04.

A desk with a quantitative team will say it does not build bands this way, and it is right. It runs quantile regression on weather features, or it buys a probabilistic forecast with bands already attached. That objection does not weaken the audit; it describes the normal case for it. **What is under test is not how a band was built but whether the number printed on it is true.** The procedure here scores whatever band you already quote, from whatever construction, against the outcomes you already have: it counts how often the band contained the answer, splits that count by regime, and prices the misses. Nothing in it needs to know the model. The two constructions audited below are in the paper because they are public and reproducible, not because they are the state of the art, and a better-built band is a more interesting subject for the same test rather than an exemption from it.

FIG. 02: THREE WAYS TO PROMISE 90%

Coverage each method actually delivered at the selected promise, per series. All three use the same trailing 90-day windows per quarter-hour slot, scored strictly one day ahead. The **madder segment** is the gap between promise and delivery. Interval width is the honest price of closing it: shown in the readout.



SAME PANEL · TRAILING 90-DAY WINDOWS PER QUARTER-HOUR SLOT · SCORED ONE DAY AHEAD · MADDER = SHORTFALL VS PROMISE

At nominal 90%, the Gaussian band delivered 84.8–88.9% depending on series; rolling conformal delivered 83.2–86.6%. Neither is a 90% band. The aggregate conceals worse: solar’s nominal-80% construction covered **56–63% through spring 2026**, precisely when new solar capacity made the error distribution grow faster than any trailing window could learn.

PITFALL 3.1: CONFORMAL IS NOT A MAGIC WORD

Split-conformal’s celebrated finite-sample guarantee is conditional on exchangeability: tomorrow’s error drawn from the same distribution as the window’s. A growing solar fleet is precisely a violation of that condition. Applied statically under distribution shift, conformal undercovers just like the Gaussian it replaced. A warranty is only as good as its conditions.

The construction that keeps its promise

Adaptive conformal inference tracks its own misses and repairs them. Coverage returns to nominal by construction.

Adaptive conformal inference (Gibbs & Candès, 2021) treats the miss rate as a control problem: after every observation, the target miss rate α_t is nudged,

$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{miss}_t),$$

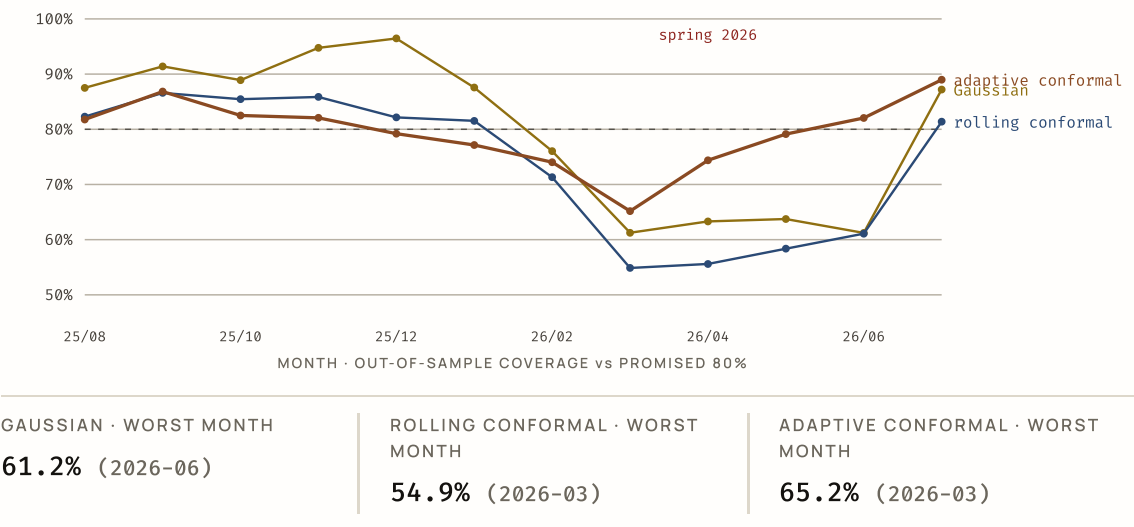
so the very next interval widens after a miss and tightens after a long run of hits. The step size γ is a plain tuning constant (larger reacts faster and wobbles more), and it was fixed at $\gamma = 0.02$ before any evaluation, not tuned to the results. Same trailing windows, same no-lookahead discipline, same data.

The result, visible in Fig. 02 for every series and level: achieved coverage lands at 89.0–89.7% against a 90% promise, 79.3–79.9% against 80%, 93.1–94.6% against 95%. The price is stated rather than hidden, and stated in the tail, not just the mean, because an average width in an article about the sins of averaging would be an irony. The adaptive load band at 90% averages 8.8 GW against the Gaussian’s 7.5 GW; at the 95th percentile of widths it quotes 14.2 GW against the Gaussian’s 11.1 GW, and its single widest interval reached 19.3 GW (onshore wind’s worst: 25.8 GW). Those wide moments are the method telling you it recently missed and does not yet trust the window. That is information, not noise. The Gaussian’s narrower quotes at those same moments *were never actually 90% bands*. Calibration does not make uncertainty larger; it stops you understating it.

Where the aggregate table still flatters everyone is regime. The month-by-month cut is the figure this article’s own Recipe demands:

FIG. 03: COVERAGE THROUGH THE REGIME BREAK

The same three constructions, scored month by month. Aggregate tables average this away; the promise is only as good as its worst regime. The shaded band marks spring 2026, when German solar additions shifted the error distribution faster than a trailing window learns.



SAME PANEL AND CONSTRUCTIONS · MONTHLY OUT-OF-SAMPLE COVERAGE · SHADED BAND = SPRING 2026 SOLAR FLEET SHIFT

The month-by-month cut also keeps us honest about what adaptivity is. When the solar shift hits in March 2026, *all three* methods get hurt: ACI’s worst month is 65% against an 80% promise, because a reactive method cannot see a regime break coming. The difference is what happens next: ACI walks back to 74% in April, 79% in May, 82% by June, while the static constructions sit in the mid-50s to low-60s for four consecutive months, still quoting “80%” the whole time. Adaptivity is not immunity; it is recovery measured in weeks instead of quarters, and a stated miss you can see happening, instead of a silent one.

RECIPE 4.1: AUDIT YOUR OWN BANDS

1. Pair each committed forecast with its realized outcome; never re-fit after the fact.
2. Score whatever band you quote (parametric or otherwise) strictly out-of-sample, per delivery slot.
3. Report coverage by regime (hour, season, fleet state), not only in aggregate.
4. If coverage drifts from the promise, adapt the miss rate, not the story.

What this audit can and cannot show

TSO forecasts are the public benchmark, not your stack. The method transfers, the numbers will not.

What do these percentage points cost? The Dutch leg answers, because it is the leg that has been run: NL imbalance prices were pulled alongside the forecasts. Running the identical audit on 13 months of NL load forecasts and pricing every escape from the trailing Gaussian “90%” band at that quarter-hour’s *adverse* imbalance-versus-day-ahead spread (favorable spreads clipped to zero: risk side only), scaled to a portfolio with 100 MW mean load whose errors move proportionally to the system’s (an assumption, stated): **€362,500 of un-reserved adverse exposure over the 13 months, against €158,000 under the adaptive band: a ≈€200k difference per 100 MW.** The stated band, incidentally, was no 90% band in the Netherlands either: it missed 15.8% of quarter-hours. And one more property the averages hide: the mean adverse spread on the intervals the Gaussian band missed was €39.4/MWh against €31.0/MWh overall. **The band fails disproportionately in exactly the intervals where failing is expensive.** Miscalibration and market stress share causes; that correlation is the audit’s sharpest commercial fact. The German reBAP leg follows on the same method; reBAP is published quarter-hourly by the four TSOs on netztransparenz.de.

Beyond that, this analysis can show whether a published point forecast is conditionally biased, and whether a band constructed around it holds its promised frequency. Those are checkable against outcomes. It cannot show whether *your* forecasts share these failure modes: portfolio forecasts differ from system forecasts, and a desk’s decision layer consumes uncertainty differently than a TSO’s.

One consequence deserves naming, because it is where this method has teeth. **If you buy forecasts from a vendor who quotes confidence bands, everything above applies to them exactly as it applies to the TSO.** Coverage is checkable by the customer, from data the customer already holds (committed forecasts and realized outcomes) without the vendor’s cooperation or their model. A vendor whose “95%” bands blow through weekly in a windy quarter is not suffering “unprecedented weather”; they are failing an audit nobody has run. Recipe 4.1 doubles as the four questions to put in your next RFP or quarterly vendor review.

The audit methodology (the pairing discipline, per-slot scoring, regime-conditional coverage, the adaptive repair) transfers unchanged to any stack that can export twelve months of forecasts and outcomes.

Forecast Calibration Audit: your committed forecasts and outcomes (or your vendor's), audited exactly as above. **Input:** one CSV per series (timestamp, forecast, actual, stated band if any); 12+ months. **You get:** the bias clock and coverage ladder for your portfolio; coverage by regime and by season; the width-tail accounting; a monetized shortfall estimate against your market's prices; an adaptive recalibration demonstration on your own data; a 45-minute readout. **Terms:** fixed price by portfolio size, two weeks from data delivery; runs on your infrastructure if preferred; your data is deleted on delivery and never used elsewhere; standard mutual NDA before anything moves. A scoped single-series **Vendor Band Audit** is available as a smaller first step; a specimen of that deliverable, run end to end on a public series, shows exactly what arrives. If your bands are honest, the audit proves that. A marketable result too.

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analysis · v1.0 ·

Code and data ↗ (<https://github.com/m-tanti/kwantil-research/tree/HEAD/analyses/german-grid-forecast-calibration>)

• Errors? Write in.